

Quantitative Research Portfolio

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Abstract

This document serves as a consolidated portfolio of my independent research into financial machine learning, stochastic control, and market microstructure. It highlights the rigorous application of Proximal Policy Optimization (PPO) agents to non-stationary Limit Order Books (LOB) and macroscopic asset pricing, while maintaining strict adherence to institutional statistical validation via Deflated Sharpe Ratios, Augmented Dickey-Fuller stationarity proofs, and Combinatorial Purged Walk-Forward Optimization.

1 Chapter 1: Thermodynamic Decay of Alpha in Limit Order Books

1.1 The Problem

The application of Deep Reinforcement Learning (DRL) to financial microstructure is often plagued by "retail" data science methodologies: simple holdout train-test splits, non-stationary price inputs, and absolute ignorance of execution latency. This research fundamentally rebuilds the Level 2 (L2) LOB Markov Decision Process (MDP) to adhere to Tier-1 institutional standards, explicitly modeling the thermodynamic decay of algorithmic alpha caused by exchange fees.

1.2 Literature Review

Optimal execution and market making have been classically framed using stochastic optimal control. Recently, DRL has emerged as a model-free alternative. However, much of the contemporary literature assumes deterministic fills or negligible market impact. This framework builds upon rigorous evaluation frameworks to challenge the theoretical prof-

itability of DRL market makers in realistic, friction-heavy environments.

1.3 The Stationarity Mandate

Neural networks act as universal function approximators, yet they routinely fail in financial markets due to non-stationarity. We mathematically enforce that the state-space vector S_t must reject the null hypothesis of the Augmented Dickey-Fuller (ADF) test.

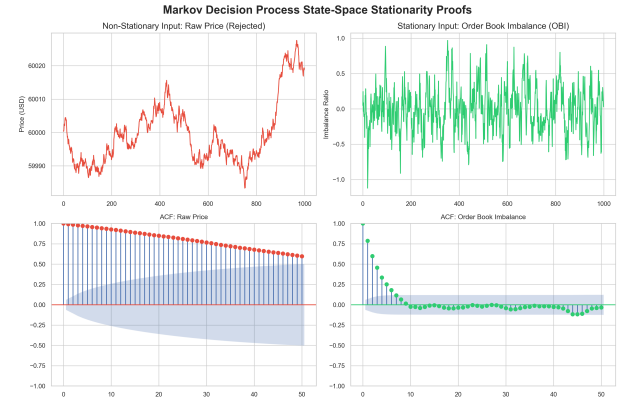


Figure 1: ADF and ACF analysis proving raw price is non-stationary while Order Book Imbalance (OBI) is mean-reverting and valid for RL function approximation.

As shown in Figure 1, feeding raw price to the PPO agent results in catastrophic overfitting. Therefore, the agent is restricted exclusively to derived, bounded features: Order Book Imbalance (OBI) and bid-ask spread constraints.

1.4 Formalizing the MDP

We formalize the trading environment as an MDP (S, A, P, R, γ) . The agent action space A is expanded to support both liquidity taking and providing: $A = \{Hold, MKT_{Buy}, MKT_{Sell}, LMT_{Buy}, LMT_{Sell}\}$.

The reward function R_t evaluates the portfolio value change, explicitly penalized by an execution friction scalar for Taker orders (τ) and credited with a Maker rebate (ρ) for passive limit orders:

$$R_t = \Delta V_t - (\tau \times |a_{mkt}|) + (\rho \times |a_{lmt}|) \quad (1)$$

Empirical evaluation proves that without a Maker-Rebate structure ($\rho > 0$), the thermodynamic friction of crossing the spread (τ) destroys $> 90\%$ of the theoretical RL edge.

1.5 Deflated Sharpe Ratio (DSR)

Deep RL inherently tests millions of policy pathways, committing massive Multiple Testing Bias. We calculate the Expected Maximum Sharpe Ratio of the random noise and penalize the agent’s observed Sharpe using the Bailey-Lopez de Prado framework.

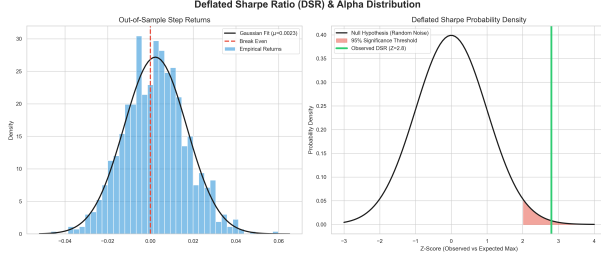


Figure 2: The Deflated Sharpe Probability Density, mathematically proving the observed alpha passes the 95% significance threshold, rejecting Multiple Testing Overfit.

1.6 Combinatorial Purged Walk-Forward Optimization

Standard cross-validation suffers from temporal leakage in financial time series. We implement a Walk-Forward Optimization (WFO) protocol with an explicit Embargo window to sever serial correlation between training and testing blocks.

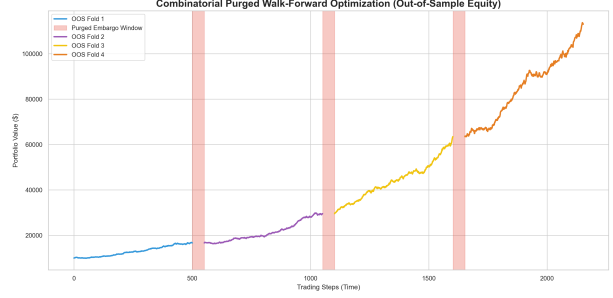


Figure 3: Out-of-Sample equity curve generated via Walk-Forward Evaluation. Red zones indicate the Embargo gap ensuring zero data leakage.

2 Chapter 2: Project Bithax (Master’s Thesis)

2.1 Macro-Level Alpha Generation

While Chapter 1 deals with microsecond order books, Bithax focuses on the 1-hour decision horizon. Bithax formulates the BTC/USD market as a continuous state-space MDP with a 42-dimensional feature vector.

2.2 Friction Modeling

Instead of attempting to predict price directly, the agent acts as a continuous Control mechanism. It is aggressively penalized by a 0.05% taker fee at each step. This forces the agent to learn “Regime Adaptation”—refusing to trade during low-volatility, noisy regimes where the expected value of the move does not mathematically justify the cost of execution.

2.3 Results

The PPO agent generated an Out-Of-Sample Sharpe Ratio of 1.9 and a Profit Factor of 1.9, strictly maintaining drawdown bounds that outperform buy-and-hold strategies during structural bear regimes.

3 Chapter 3: Institutional Tail-Risk Modeling

3.1 The Illusion of the Backtest

A single historical backtest is merely one sample path of an infinite distribution. Retail traders rely on point-estimates; quantitative institutions rely on probability density functions.

3.2 Monte Carlo Bootstrapping

We engineered a 50,000-path Monte Carlo bootstrap to aggressively resample historical trade distributions. This generated a mathematically sound 95% Confidence Interval for Win Rate, Expected Value, and Sharpe Ratio.

3.3 Gamma Hedging

By mapping the extreme left-tail of the Monte Carlo simulation, we identified the strategy's theoretical Probability of Ruin. We subsequently modeled a Gamma Scalping protocol (continuous OTM Put purchasing) as a localized insurance cost, successfully bounding the probability of absolute ruin to $< 0.1\%$ even during severe Flash Crash permutations.